# Reliable Probabilistic Time Series Imputation

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## Shared research problem

**Motivation:** plausible reconstructions and low median error can hide unreliable uncertainty. Narrow data, leakage and unsuitable metrics can distort evaluation (Bergmeir, 2023).
**Task:** probabilistic retrospective imputation on ETT_h1, Electricity and BeijingAir. Compare scattered points and contiguous gaps at equal hidden counts.

## Project question

**Question:** when are native predictive distributions reliable, and how does two-sided forecasting adaptation affect calibration?
**Required methods:** Chronos-2-small, conditional diffusion CSDI via PyPOTS, and a Gaussian Kalman smoother.
**Hypotheses:** blocks are harder than scattered points; median accuracy and calibration can disagree; blending can change calibration in either direction.

## Assumptions, methods and evaluation protocols

| | CSDI via PyPOTS | Gaussian Kalman smoother | Chronos-2-small |
|---|---|---|---|
| **Model role** | Trained conditional diffusion imputer | Classical probabilistic state-space baseline | Frozen quantile foundation forecaster |
| **Input and fit** | Same masked window; pooled univariate training on point/block mixture | Condition on all observed positions; fit process and observation variances on train | Left context and reversed right context; other hidden runs remain masked |
| **Native uncertainty** | Sample missing trajectories; retain 100 draws per window if feasible | Conditional Gaussian distribution of each missing measurement | Native forecast quantiles; two-sided blend is our adaptation |
| **Point-error score** | Empirical sample median | Median of the conditional Gaussian | Median of left-only, right-only and blended quantiles |
| **Primary check** | Conditioning masks; sample extraction; 50/100-draw sensitivity | Include observation variance; inspect seasonal misspecification | Matched left/right/blend ablation; gap-position coverage and width |
| **Claim boundary** | **Finite draws introduce quantile error; compute must pass pilot** | **Gaussian assumptions may fail; fitted-parameter uncertainty excluded** | **Forecast calibration does not imply imputation calibration** |

## Evaluation metrics

| Metric | Formula | Purpose and interpretation |
|---|---|---|
| **Classical point metrics** | | |
| **MAE / MSE / RMSE** | MAE = mean \|y − m\|<br>MSE = mean (y − m)²; RMSE = √MSE | Point accuracy of the median m. Lower is better. |
| **MRE** | Σ \|y − m\| / Σ \|y\| | Relative point error in original units. Guard near-zero denominators. |
| **Probabilistic scores and uncertainty diagnostics** | | |
| **Pinball loss** | ρq(e) = max(qe, (q − 1)e), e = y − Qq | Quantile accuracy. Average over q = 0.1, …, 0.9. Lower is better. |
| **CRPS** | E\|X − y\| − ½ E\|X − X′\| | Marginal distribution accuracy.<br>Kalman: analytical. CSDI: samples. Chronos: approximation. |
| **NLL** | −log p(y \| observed) | Predictive density at the truth. Kalman: exact.<br>Samples or quantiles alone are insufficient. |
| **PCE** | mean_q \|q − c(q)\|; c(q) = mean 1[y ≤ Qq] | Quantile calibration error. Zero is ideal. Inspect reliability curves too. |
| **CCE** | mean_s [s − coverage(s)] | Signed interval calibration. Positive: undercoverage. Negative: overcoverage. |
| **80% coverage** | mean 1[Q0.1 ≤ y ≤ Q0.9] | Calibration of the central 80% interval. Target: 0.8, alongside interval width. |
| **Width / SIW** | Width_s = Q[(1+s)/2] − Q[(1−s)/2]<br>SIW = mean_s {mean(Width_s) / IQR_s(y)} | Sharpness. IQR_s is the matching empirical target range. Read with coverage. |

## Claims vs evidence

| Source | Relevant evidence | Project use | Not established |
|---|---|---|---|
| **Bergmeir 2023** | Plots, leakage, weak benchmarks and unsuitable metrics can mislead | Use matched masks and strong simple baselines | His article does not evaluate probabilistic imputation |
| **TSI-Bench 2024** | Performance depends on dataset and missingness pattern | Standard datasets; point/block comparison; MAE, MSE, MRE | Published scores are not reproduced by our changed protocol |
| **Adler et al. 2026** | Calibration requires diagnostics beyond point accuracy | PCE, CCE, SIW and reliability curves | Forecast calibration does not validate a two-sided blend |

Matched comparison and decisive ablation
Masks: 24 or 128 hidden values, scattered or contiguous. Same hidden count within each pair; both window edges stay observed.
Comparator: Gaussian smoothing conditions on all observed positions. Evaluate uncertainty for missing measurements, not only latent states.
Ablation: left-only vs right-only vs blended Chronos quantiles on identical targets. Plot coverage and width along the gap.
Sampling check: validate CSDI quantile stability with 50 vs 100 draws. The Gaussian figure is analytical; no benchmark results yet.

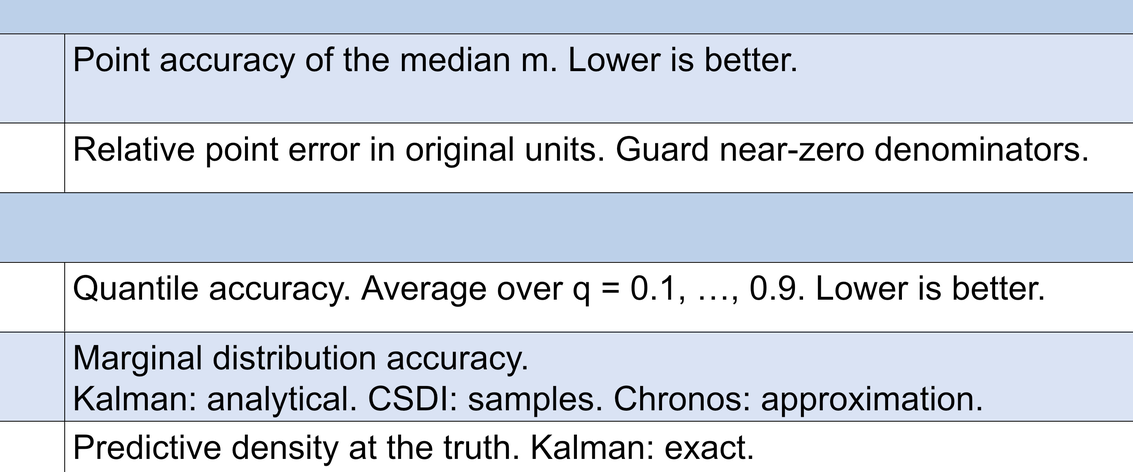



Fig. 1. Gaussian random walk with noiseless endpoints and a 24-step gap. Near the midpoint, blended quantiles are about 1.4 times wider than the exact bridge. Analytical illustration, not measured TSFM performance.

## Work plan

| Week 1: Data, masks and CSDI feasibility; Chronos adapters | Week 2: Gaussian baseline, pilot and small CSDI training | Week 3: Freeze models and draws; matched evaluation | Week 4: Calibration diagnostics; poster and report draft | Week 5: Rebuild outputs, reproduce and peer review | Week 6: Final report and limitations |
|---|---|---|---|---|---|

Protocol: chronological 60/20/20% train/validation/test; hidden targets only; train-only scaling. Medians provide point-error scores.
Scoring: MAE, MSE, guarded MRE; PCE, CCE, SIW, reliability and coverage; pinball loss; CRPS and NLL where evaluable (lower is better).
Compute: Chronos MPS smoke test, CPU/CUDA fallback. Required CSDI: early feasibility check and six GPU-hours total; reduce shared scope if needed.
Limits: synthetic masks; eligible complete windows; dependent channels; pretraining overlap; finite diffusion samples; Gaussian misspecification.
References: Bergmeir 2023; Du et al. 2024; Adler et al. 2026; Tashiro et al. 2021; Gneiting & Raftery 2007; PyPOTS and Chronos.